CURRICULUM VITAE

HANNAH K. KOVIC

GEOLOGIST

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PRIMARY RESEARCH INTERESTS

Structural geology and tectonics, tectonic geomorphology, mountain-building processes, geochronology, thermochronology, deformation rates, exhumation history, landscape and topographic evolution.

NATIONALITY—PA, UNITED STATES
CURRENT RESIDENCE—VD, SWITZERLAND



EDUCATION

May 2025 — **B.S. Geology** *with Honors* | University of Pittsburgh | Pittsburgh, PA, USA
Cumulative GPA 3.679 / 4.00 | Geology GPA 3.789 / 4.00
Thesis Discipline & Advisor: *Tectonics & Structural Geology*, Dr. Nadine McQuarrie

Jun 2020 — **High School Diploma** *High Honors Distinction*
Unionville High School | Kennett Square, PA, USA

AWARDS & DISTINCTIONS

May 2025 — ***Magna Cum Laude*** Honors status achieved upon graduation with a GPA >3.5

May 2022 — ***Departmental Honors*** GPA > 3.25, research credits >3, write and defend a graduate-equivalent thesis.

Dec 2023–May 2025 — ***Dean's List*** Awarded by semester. Minimum 12 letter grade credits with a term GPA of at least 3.50

ACADEMIC GRANTS

May 2024 — ***Norman K. Flint Field Scholar Award*** $1700 USD
Academic grant awarded to field geologists based on academic merit.

Oct 2022 — ***Pitt Green Fund Sustainability Grant*** $100 USD
Plantsburgh Sustainability Project: Increased greenery on the University of Pittsburgh Campus, hosted an educational event, and presented findings at the annual Pitt Sustainability Symposium

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RESEARCH EXPERIENCE

Aug 2023—Present — **University of Pittsburgh, Department of Geology & Environmental Science *Research Assistant***

- Data compilation and projection (*ArcGIS Pro, Python, Microsoft Suite*)
- Hypothesized and tested four balanced cross section geometries and kinematics (*Move PetEx*)
- 2D Flexural kinematic forward modeling flexural isostacy and erosion (*Move PetEx*)
- 2D Thermokinematic advection diffusion modeling thermal cooling history (*Pecube-D, MATLAB*)

CONFERENCES & LITERATURE

H. Kovic, N. McQuarrie, S. Kosowsky, *Hypothesizing Taiwan's subsurface structural geometries and evolution using flexural-kinematic and thermochronometric modeling techniques* (Thesis Paper). Publication in progress.

H. Kovic, Nadine McQuarrie, P-C Chiang, S. Das, K.-M. Johnson, B.-J Yanites, *Structural drivers for fault-driven topographic uplift in central Taiwan.* ID#1598474. American Geophysical Union Fall Meeting, December 2024, Washington DC. (Abstract & Poster).

LAB EXPERIENCE

Sep 2025—Present — **University of Lausanne, Institute of Earth Surface Dynamics *Lab Intern***

- ESR, OSL, and IRSL trap charge thermochronometry performed on 14 rock samples to constrain Taiwan's tectonic evolution and exhumation history.
- Cutting, crushing, sieving, chemical treatment, SPT density separation, magnetic separation (*FRANTZ*)
- ESR experiments (*Irradiation SAR-Additive & Annealing ITD*), ESR measurements (*Magnet Tech*)
- OSL & IRSL experiments (*Riso*), OSL & IRSL measurements (*Riso*)
- Data plotted and analyzed (*MATLAB, Microsoft Suite*)
- Publication in Progress

FIELD EXPERIENCE

Aug 2025 **U.S. Geologic Survey, Washington State (USA) Field Excursion—*Field Assistant***
2-week field investigation. Earth Mapping Resources Initiative (Earth MRI). Magnetic and radiometric survey of Northeastern Washington State. Geologic and structural mapping, collected oriented samples for thermochronology and magnetic susceptibility, and geophysical gravity survey (*Gravimeter*).

Mar 2025 **University of Pittsburgh, Taiwan Field Excursion—*Field Investigator***
2-week field investigation. Geologic and structural mapping, collected samples for ESR, OSL, and IRSL thermochronometry. Structural data collection and projection using paper maps and ArcGIS Pro. NSF-funded.

May 2024—Jun 2024 **Idaho State University—*Geology Field Study***
5-week field course in Idaho. Applied skills in field surveying, structural mapping, GIS and remote sensing, hydrology, sedimentology, and igneous metamorphic petrology. Full letter of acquired skills provided upon request.

May 2023 **University of Pittsburgh—*Himalaya Field Study***
4-week field course in the Indian Himalayas. NOLS outdoor leadership certification, Wilderness Advanced First Aid Certification, CPR Certification, Leave No Trace Certification.

TEACHING EXPERIENCE

Aug 2024—Dec 2024 **University of Pittsburgh Lab Assistant**—***Mineralogy***
Conducted hand sample and thin section identification, sample organization, and thin section preparation with rock saws for the Mineralogy Lab course.

Aug, 2024—Dec 2024 **University of Pittsburgh Lab Teaching Assistant**—***Sedimentology & Stratigraphy***
Assisted with lab instruction, taught field skills in sedimentology, stratigraphic measurement (with Jake Staff), sedimentary rock identification and classification, soft rock structural identification, and historical geologic analysis.

Jan 2024—May 2024 **University of Pittsburgh Lab Teaching Assistant—*History of Earth***
Assisted with lab instruction, taught geologic field methods, sedimentary stratigraphy, rock and mineral identification, and historical geologic analysis.

SUMMARY OF TECHNICAL SKILLS & CERTIFICATIONS

Data Projection & Analysis	*ArcGIS Pro, MATLAB, Microsoft Suite*
Modeling	*Pecube-D, Move PetEx, MATLAB*
Laboratory Methods	*ESR (SAR, ITD, Magnet Tech); OSL & IRSL (Riso); FRANTZ, Thin Section Analysis*
Data Acquisition	Geophysical Gravity Survey *(Gravimeter),* Structural mapping *(Brunton)*
Certifications	*Wilderness Advanced First Aid (WAFA), CPR, NOLS Outdoor Leadership*

PROJECTS & ENGAGEMENT

Aug 2024—Present American Geophysical Union *Member*
Dec 2023—May 2025 Pittsburgh Geological Society *Member*
Oct 2023—Present Geologic Society of America *Member*
Aug 2023—May 2025 University of Pittsburgh Geomorphology and Tectonics Research Reading Group *Member*
Oct 2022—May 2025 University of Pittsburgh Outdoors Club *Member*
Dec 2021—May 2022 Mailroom Recycling Sustainability Project *Project Lead*
Aug 2021—Dec 2021 Plantsburgh Sustainability Group *Founder*